

Scientific Research and Reproducibility Standard

Requirements for experiments, notebooks, models, instruments, and computational work

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Under Review. This first-edition record is complete for institutional review but does not become current authority until approved and published.

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1. Purpose

This standard governs scientific and engineering research records created, imported, or published through Sustainable Catalyst. It supports reproducibility, methodological review, safety, and durable institutional memory.

2. Research questions and protocols

A research record should identify the question, rationale, hypothesis or objective, method, variables, controls or comparisons, success criteria, expected limitations, and safety requirements before results are interpreted.

3. Notebook integrity

Notebooks should preserve dates, authorship or responsible operator, materials, instruments, software, environmental conditions, parameter changes, observations, deviations, failures, and links to datasets and outputs. Corrections should remain distinguishable from original entries.

4. Data provenance

Datasets should record origin, collection method, units, schema, coverage, licensing, transformations, exclusions, missing-data treatment, and version. Raw data should be preserved where rights, privacy, safety, and storage permit.

5. Instruments and calibration

Instrument records should identify device, model, configuration, calibration or verification status, accuracy or resolution where relevant, firmware or software, and environmental conditions. Simulated instruments must not be presented as physical measurements.

6. Computational environments

Computational research should preserve code version, language and package versions, environment or container information, hardware constraints where material, random seeds, parameters, input identifiers, and execution logs sufficient for review.

7. Models and solvers

Model equations, boundary conditions, assumptions, numerical methods, convergence criteria, solver tolerances, validation data, and sensitivity should be documented. Numerical convergence does not establish scientific validity.

8. Validation

Validation may include analytical solutions, benchmark datasets, independent implementations, controls, calibration standards, dimensional checks, residual analysis, error bounds, or comparison with observed outcomes. The validation method should match the claim.

9. Reproduction and replication

Reproduction repeats an analysis using the same data and method. Replication tests whether a finding holds with independent data, implementation, instrument, or context. Records should not claim replication when only a rerun was performed.

10. Uncertainty and significant figures

Reported precision should reflect measurement and model limitations. Uncertainty sources should be identified. Significant figures, error bars, confidence intervals, sensitivity ranges, and qualitative uncertainty should be used appropriately rather than decoratively.

11. Negative, null, and inconclusive results

Failed experiments, null results, unstable models, nonconvergence, and contradictory observations are legitimate research records. They should not be erased merely because they do not support the initial hypothesis.

12. Safety and ethical boundaries

Laboratory, field, hardware, biological, chemical, electrical, mechanical, aerospace, medical, and environmental work may create hazards. Public examples are educational and do not replace risk assessment, supervision, protective equipment, regulatory compliance, or qualified professional review.

13. Human and sensitive data

Research involving people, identifiable information, protected groups, health information, or sensitive locations requires appropriate consent, privacy, security, minimization, and institutional or legal review. Public tools should not be treated as an ethics review process.

14. AI and automation

AI may assist code, literature review, classification, protocol drafting, or anomaly detection. It must not fabricate data, observations, citations, calibration, validation, or experimental completion. Automated runs require logs, bounded permissions, and human review.

15. Publication and bundles

A reproducible research bundle should include a readable summary, protocol, data references, code or method, environment, parameters, outputs, validation, limitations, license information, and integrity hashes where practical.

16. Handoffs

Lab artifacts sent to Workbench, Site Intelligence, Knowledge Library, or Decision Studio should retain experiment identity, conditions, data version, method, validation state, and unresolved limitations.

17. Review

Research claims should be reviewed at a level proportionate to consequence. Educational demonstrations, exploratory notebooks, validated methods, and professional analyses must not be presented as equivalent states.

Related Foundation Documents

- SC-FND-003
 - SC-FND-005
 - SC-FND-006
 - SC-FND-011
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Revision History

Version	Date	Status	Summary
1.0.0	2026-07-16	Under Review	Institutional Foundations First Edition draft prepared for review and publication.

Authority statement. This living HTML document is the proposed first-edition record within its defined scope. Fixed PDF editions preserve review snapshots. Earlier records remain available for historical reference.